Highlights

Multi-Agent Reinforcement Learning for Surface–Subsurface Observation of Tropical Cyclones Using Submersible Unmanned Vessels

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- Dual-mode SUVs track surface winds and sample the subsurface environment.
- Event-specific ERA5–HYCOM environments cover 500 historical TCs.
- Parameter-sharing IPPO algorithm scales TC tracking across 4–16 SUVs.
- Current and future SUV configurations are compared for surface and subsurface observation.

Multi-Agent Reinforcement Learning for Surface–Subsurface Observation of Tropical Cyclones Using Submersible Unmanned Vessels

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ABSTRACT

Tropical cyclones are destructive, while intensity and rapid-intensification forecasts remain limited by sparse near-core observations. Traditional observation platforms rarely combine rapid transit, surface-wind sampling, and subsurface-current sensing, while Blue Whale-class submersible unmanned vessels can transit rapidly at the surface to reach TC core regions and then dive to observe temperature, salinity, currents, and biogeochemical properties at multiple depths. To enable autonomous environmental perception and decision making, we adopt a simulation-based multi-agent reinforcement-learning framework. For training and evaluation, we construct event-specific environments that combine ERA5 10 m winds, HYCOM currents from 0 to 60 m, and IBTrACS tracks for 500 western North Pacific TCs. A parameter-sharing independent proximal policy optimization (IPPO) policy maps shared surface-wind observations or local current histories to continuous commands every 10 min. We compare 12 m/s surface and 1.5 m/s submerged limits, close to current Blue Whale design speeds, with 20 m/s and 2.0 m/s as potential future upgrades. At the surface, the fleet follows the TC toward high-wind regions despite uncertain deployment, reducing the final tracking error and improving observation benefit. Underwater, across all seven depths, active control consistently outperforms random control and passive drift, achieving a higher mean experienced current speed, longer trajectories, and broader spatial coverage. The framework provides a learning-based, closed-loop control methodology for future coordinated surface–subsurface TC observation with SUVs.

1. Introduction

Tropical cyclones (TCs) threaten coastal communities, infrastructure, and offshore operations, and the proportion of intense and rapidly intensifying events has increased in many basins [1, 2, 3, 4]. Landfalling TCs over the western North Pacific and China have also become more destructive [5, 6, 7, 8]. Track prediction has advanced through numerical weather prediction, satellite assimilation, ensembles, and data-driven models, whereas intensity prediction remains more difficult [9, 10]. A central reason is that intensity depends on air–sea enthalpy and momentum exchange, upper-ocean heat content, barrier layers, mesoscale eddies, and TC-induced currents [11, 12]. Yet clouds and rain limit microwave sensing, scatterometers saturate at high winds, and spaceborne sensors cannot resolve the subsurface ocean [13, 14, 15]. Near-core wind and upper-ocean states therefore remain poorly constrained in many forecast initial conditions.

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Existing observing platforms each cover only part of this problem. Satellites provide broad coverage but cannot simultaneously observe the surface wind maximum and three-dimensional upper ocean [13]; aircraft obtain high-quality near-core measurements but are expensive and campaign-limited [16, 17]; and gliders can remain at sea for long periods but move much more slowly than a translating TC [18]. AUVs can adapt their sampling to changing environmental and vehicle states [19]. Upper-ocean observations can improve initial states and, in some cases, intensity forecasts [20, 21, 22, 23, 24], but their value is highly sensitive to sampling location and background structure [25, 26, 27]. Adaptive TC observation is therefore a sequential decision problem involving motion, sampling time, energy, and coordination rather than coverage alone.

A submersible unmanned vessel (SUV) combines rapid surface transit with subsurface sensing [28, 29, 30, 31]. The Blue Whale is used as an engineering reference: its public specifications indicate a surface speed of about 23 kn (11.83 m/s), a submerged speed of about 3 kn (1.54 m/s), and a reference depth near 60 m [28]. Rather than claiming a digital twin, this study parameterizes identifiable future Blue Whale-class capabilities. In the proposed mission, the surface fleet observes 10 m winds, follows the TC toward the high-wind region as closely as possible, and is evaluated by its final tracking error. At depth, four SUVs are assigned

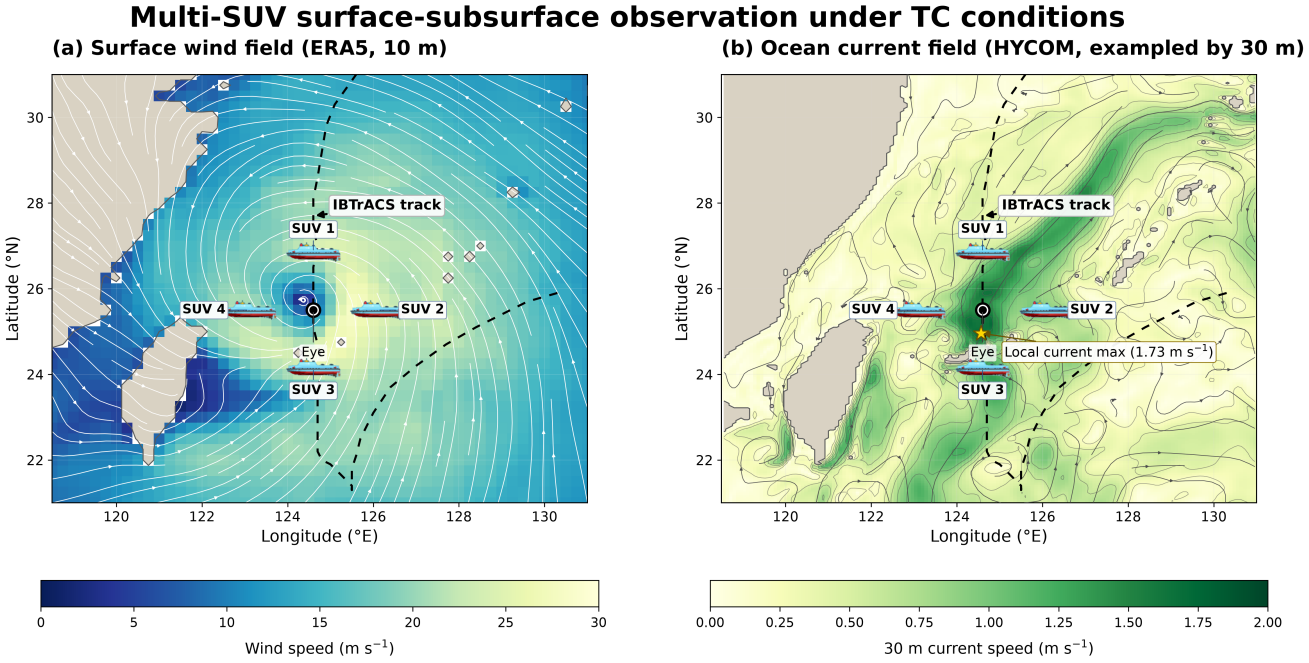


Figure 1: Coordinated surface–subsurface TC observation for a representative HINNAMNOR (2022) snapshot. The left panel shows four SUVs tracking the IBTrACS best track while sampling ERA5 winds at 10 m above the sea surface. The right panel shows the same formation sampling HYCOM currents at 30 m below the sea surface; this depth is shown as an illustrative example, whereas the complete experiments cover the seven depths from 0 to 60 m. The four-vehicle configuration is illustrative, with equal angular spacing; subsequent experiments evaluate fleets of 8, 12, and 16 vehicles.

to each prescribed layer and seek the strongest observable current. Figure 1 illustrates this handoff; four vehicles are shown at the surface as an example, and the same parameter-sharing policy is later evaluated with 4, 8, 12, and 16 SUVs.

Classical graph search, sampling-based planning, potential fields, and model predictive control are effective when the cost map and dynamics are known [32, 33, 34, 35, 36, 37]. In a TC, however, the fields are nonstationary and only locally observed, while propulsion is superimposed on ambient flow. Conventional pipelines also estimate a field or waypoint sequence before execution. By contrast, reinforcement learning (RL) maps each new measurement directly to the next action, closing the observation–decision–motion loop. RL has demonstrated strong sequential decision making across diverse domains [38, 39, 40, 41, 42, 43, 44, 45], and independent PPO is competitive in cooperative multi-agent tasks [46, 47, 48]. Multi-AUV adaptive sampling and target tracking provide the ocean-robotics foundations [49, 50, 51, 52, 53].

Three gaps remain. First, prior TC observation studies often represent the ocean with analytical or reduced-order flows rather than event-specific fields [54, 55]. Second, track-forecast errors require rapid, adaptive transit from an uncertain deployment point to the high-wind region [25, 26, 27]. Third, the closest multi-AUV active-observation study used only four vehicles [55]. We therefore formulate stochastic policy optimization under partial observability [56, 46, 47], extend the earlier center-inference ideas [54, 55] to event-specific surface–subsurface observation, and evaluate 4-, 8-, 12-, and 16-SUV fleets.

The main contributions of this study are summarized as follows.

1. We construct an event-specific learning environment for 500 TCs by aligning ERA5 10 m winds, HYCOM currents at seven depths from 0 to 60 m, and IBTrACS tracks.
2. Wind-only parameter-sharing IPPO learns inward motion from training-time center rewards and lowers the final tracking error, measuring how closely the surface fleet follows the TC and enters the high-wind region.
3. We compare active IPPO, random control, and passive drift for subsurface observation with four SUVs at each depth, using the highest attainable observed current speed as the objective.

2. Problem Formulation

2.1. Observation Task and Design Requirements

The objective is to coordinate a fleet of submersible unmanned vessels (SUVs) to observe a moving tropical cyclone (TC) across the sea surface and upper ocean. At the surface, each SUV navigates toward the predicted TC center and high-wind region, with the final tracking error used to evaluate how closely the fleet follows the TC. After entering a predefined high-wind observation window, four

SUVs descend to each prescribed depth and seek the highest attainable current speed. The framework is designed to satisfy four operational requirements:

1. rapid correction of initial deployment errors using surface propulsion;
2. decentralized tracking of the moving TC using onboard wind observations and inter-vehicle observation sharing;
3. subsurface observation at ocean depths of 0, 10, 20, 30, 40, 50, and 60 m below the sea surface;
4. safe operation under action-smoothness and coastline constraints.

The surface maximum speed is varied between 12 and 20 m/s, while the submerged maximum propulsion speed is 1.5 or 2.0 m/s. This difference creates an asymmetric decision problem: surface propulsion is used to follow the TC and reach the high-wind region as closely as possible, quantified primarily by the final tracking error. At depth, the objective changes to finding and sampling the highest possible current speed, so the four SUVs must weigh active propulsion against passive advection by the ambient current.

2.2. Problem Definition

The fleet contains N SUVs, where $N \in \{4, 8, 12, 16\}$, indexed by $i \in \{1, 2, \dots, N\}$. The position, propulsion velocity, and operating mode of SUV i at time t are denoted by $\mathbf{x}_i(t)$, $\mathbf{p}_i(t)$, and $m_i(t) \in \{\mathrm{s}, \mathrm{u}\}$, respectively, where s and u denote the surface and subsurface modes. The propulsion speed satisfies

$$\|\mathbf{p}_i(t)\| \le \begin{cases} V_{\max}^{\mathrm{s}}, & m_i(t) = \mathrm{s}, \\ V_{\max}^{\mathrm{u}}, & m_i(t) = \mathrm{u}. \end{cases} \tag{1}$$

Each simulated episode is assigned to one stage, consistent with the separate surface and subsurface environments used in training and evaluation. Thus, $m_i(t)$ selects the applicable speed bound and motion law for that episode; continuous diving, ballast transitions, and resurfacing are not simulated within a single decision trajectory.

The horizontal motion of the SUV is modeled as

$$\dot{\mathbf{x}}_i(t) = \begin{cases} \mathbf{p}_i(t), & m_i(t) = \mathrm{s}, \\ \mathbf{p}_i(t) + \mathbf{u}_{\mathrm{ocean}}\left(\mathbf{x}_i(t), z_i, t\right), & m_i(t) = \mathrm{u}, \end{cases} \tag{2}$$

where $z_i \in \{0, 10, 20, 30, 40, 50, 60\}$ m and $\mathbf{u}_{\mathrm{ocean}}(\cdot)$ is the HYCOM current at the corresponding position and depth. Thus, the underwater ground velocity is the superposition of active propulsion and background advection.

For numerical implementation, the continuous model is integrated explicitly at the decision interval $\Delta t = 10$ min:

$$\mathbf{x}_{i,k+1} = \begin{cases} \mathbf{x}_{i,k} + \mathbf{p}_{i,k}\Delta t, & m_{i,k} = \mathrm{s}, \\ \mathbf{x}_{i,k} + \left[\mathbf{p}_{i,k} + \mathbf{u}_{\mathrm{ocean}}(\mathbf{x}_{i,k}, z_i, t_k)\right]\Delta t, & m_{i,k} = \mathrm{u}. \end{cases} \tag{3}$$

The current field is evaluated at the beginning of each interval and interpolated to the current vehicle position. For the active underwater reward, we write this pre-move current as $\mathbf{u}_{i,k} = \mathbf{u}_{\mathrm{ocean}}(\mathbf{x}_{i,k}, z_i, t_k)$.

This equation is deliberately a reduced-order horizontal point-mass kinematic–advection model rather than a six-degree-of-freedom rigid-body dynamic model. It does not resolve added mass, hydrodynamic drag, roll, pitch, yaw or heading dynamics, buoyancy and ballast dynamics, thruster transients, wave loads, or actuator delay. This simplification is consistent with the information content of the environmental forcing. ERA5 winds are available at 0.25° and hourly resolution, whereas HYCOM currents are available at approximately 0.04° by 0.08° and three-hourly resolution. These fields are interpolated to the 10 min decision interval and to each vehicle position, but such interpolation does not recover high-frequency wave forcing, local wake effects, or unobserved vehicle attitude and actuator states. Introducing a detailed six-degree-of-freedom model would therefore require platform-specific hydrodynamic coefficients and forcing information that are not commensurate with the present data, and could imply a level of numerical precision unsupported by the inputs. The reduced model retains the first-order effects relevant to mission-level planning: mode-dependent speed limits, propulsion commands, ambient-current advection in subsurface mode, and coastal constraints. Accordingly, the results are interpreted as mission-planning and coordination evidence rather than as predictions of attitude, structural loading, or low-level controller response. Coupling this planning layer to an experimentally calibrated six-degree-of-freedom vehicle model is left for future work.

The local observation of SUV i is

$$\mathbf{o}_i(t) = \left[\mathbf{s}_i(t), \mathbf{v}_i(t), \mathbf{h}_i(t)\right], \tag{4}$$

where $\mathbf{s}_i(t)$ contains the observation used in the corresponding stage and $\mathbf{v}_i(t)$ is the normalized commanded propulsion velocity, $\mathbf{v}_i = \mathbf{p}_i / V_{\max}^{m_i}$; it is not the total ground velocity. In the final surface experiments, $\mathbf{s}_i(t)$ consists only of the normalized wind patches at 10 m above the sea surface sampled by the N SUVs, and no historical frame is used. Surface operation assumes idealized, lossless, and instantaneous inter-vehicle communication: every SUV receives the wind observations acquired by all other SUVs in the fleet as part of its policy input. In the underwater active-control comparison, no inter-vehicle communication is assumed. Accordingly, $\mathbf{s}_i(t)$ consists only of SUV i's local current speed and components at its assigned ocean depth, while $\mathbf{h}_i(t)$ contains a five-frame history of that same vehicle's local current. Each underwater SUV therefore knows neither the current observations nor the positions or actions of the other SUVs. The complete reanalysis field, the global best-track center, and future environmental states are not provided to the policy. Accordingly, the surface observation dimension is $d_o^{\mathrm{s}} = 3N + 2$. The underwater active controller uses a 1×1 local-current observation plus a five-frame history

and two normalized propulsion components, giving $d_o^{\mathrm{u}} = 5 \times 3 + 2 = 17$.

During training, the reward function can access the distance between each SUV and the best-track TC center. This distance supervises the association between local 10 m wind patterns and proximity to the TC center/high-wind region; the center coordinate itself is not an actor input. During validation, rewards do not influence actions. The frozen policy receives only the shared fleet-level wind samples and must track an unseen TC, thereby testing whether the learned wind-to-center relation generalizes.

The normalized action is

$$\mathbf{a}_i(t) \in [-1, 1]^2, \tag{5}$$

and is projected onto the unit disk before being mapped to propulsion in both modes:

$$\tilde{\mathbf{a}}_i(t) = \frac{\mathbf{a}_i(t)}{\max\left(1, \|\mathbf{a}_i(t)\|_2\right)}, \qquad \mathbf{p}_i(t) = V_{\max}\tilde{\mathbf{a}}_i(t). \tag{6}$$

This mapping ensures that the total propulsion speed never exceeds the mode-specific speed limit.

The cooperative task is formulated as a decentralized partially observable Markov decision process (Dec-POMDP):

$$\mathcal{M} = \langle \mathcal{I}, \mathcal{S}, \{\mathcal{O}_i\}, \{\mathcal{A}_i\}, P, R, \gamma \rangle, \tag{7}$$

where $\mathcal{I}$ is the set of SUVs, $\mathcal{S}$ is the joint environment state, $\mathcal{O}_i$ and $\mathcal{A}_i$ are the local observation and action spaces of vehicle i, P is the transition probability, r_i is the local reward contribution, $R = (1/N)\sum_{i=1}^{N} r_i$ is the shared team reward, and γ is the discount factor.

3. Data and Simulation Environment

3.1. Reanalysis and Best-Track Products

ERA5 supplies hourly zonal and meridional wind components at 10 m above the sea surface on a regular 0.25° grid [1] [57]. Wind speed and direction are calculated from u_{10} and v_{10}. HYCOM supplies three-hourly zonal and meridional currents at ocean depths of 0, 10, 20, 30, 40, 50, and 60 m below the sea surface [2] [58]. The native horizontal grid is approximately 0.04° in latitude by 0.08° in longitude. IBTrACS provides TC time, center position, maximum wind, minimum pressure, and related metadata at mostly 3 or 6 h intervals [3] [59]. The common study domain is 98.7°E–171.6°E and 6.8°N–56.7°N. The overall event set spans 19 November 2007 to 3 December 2025 UTC. Thus, the surface observation variable is the 10 m wind speed and direction, whereas the subsurface observation variables are the horizontal current velocity, represented by `water_u` and `water_v`, and the derived current speed over the 0–60 m layer. Table 1 summarizes the products, variables, native grids, and temporal resolutions used to construct the event-specific environments.

[1] ERA5 download portal: https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels.

[2] HYCOM GLBy0.08 archive: https://www.hycom.org/data/glby0p08/expt-93pt0; HYCOM ESPC-D-V02 archive: https://www.hycom.org/data/espcd-v02.

[3] IBTrACS download portal: https://www.ncei.noaa.gov/products/international-best-track-archive.

3.2. Spatiotemporal Alignment and Track Segmentation

All fields are mapped to a common simulation clock. Surface atmospheric fields are linearly interpolated from hourly values, and ocean currents are linearly interpolated from three-hourly values to the 10 min decision interval. This is an offline reanalysis simulation: adjacent reanalysis samples are used to reconstruct the field at the simulated time, and the policy receives only the interpolated local value at that time, not a future time index or full future field. This construction does not emulate the data latency of a real-time forecast system. At each decision time, the observation operator uses the reconstructed field, while the best-track center is used only for training supervision and evaluation. Any NaN or infinity produced by interpolation or coastal masking is replaced by a pre-specified neutral value after finite-difference gradients are computed. The resulting observations are checked for finiteness before entering the policy.

Although the native grids and sampling intervals are much coarser than the 10 min decision step, the environment performs online spatial and temporal interpolation at every decision time. Each SUV queries the interpolated field at its current position and at the corresponding simulation time, and the same interpolated values are used for its observation and for current-induced motion. These values are treated as the best available estimates from the archive under the stated interpolation scheme, rather than as resolved high-resolution truth or statistically optimal reconstructions of unobserved sub-grid variability.

Policy training and evaluation are offline with respect to the environmental data: the ERA5 and HYCOM trajectories are precomputed, and vehicle motion does not feed back into the wind or current fields. This one-way forcing assumption is motivated by the large separation of scales between a small SUV and a TC-scale atmosphere–ocean system, for which the momentum and energy transferred by the vehicles are negligible relative to those driving the environmental fields. It differs from an online two-way coupled simulation in which actions modify the environment and the modified environment then changes subsequent observations. The term offline here does not mean batch or offline RL: the policy is still trained through sequential closed-loop interaction with the precomputed environment, but the environmental trajectory itself is independent of the policy. A two-way coupled simulation would be required if vehicle-induced wakes or other local environmental feedback were the object of study.

The 10 min decision interval is selected as a compromise between control responsiveness and the information content of the source products. A longer interval would reduce

Table 1
Data used to construct the TC observation environment. The dataset integrates ERA5 surface winds, HYCOM currents at seven depths, and IBTrACS best tracks.

Data product	Source	Variables used	Native grid	Native time
Surface wind at 10 m above sea level	ERA5	u_{10}, v_{10}, speed, direction	$0.25° \times 0.25°$	Hourly
Current at 0–60 m ocean depth	HYCOM	water_u, water_v, speed at 0/10/20/30/40/50/60 m	$\sim 0.04°$ lat $\times 0.08°$ lon	3-hourly
Best track	IBTrACS	Position, time, wind, pressure	Point track	Mainly 3/6-hourly

Note: although the SUV platform can carry sensors for observing temperature, salinity, and currents, the reinforcement-learning policy uses only surface wind and subsurface current observations as environmental inputs.

the frequency of trajectory correction and degrade decision precision. A substantially shorter interval would increase decision and execution overhead and would substantially increase the number of decisions per episode, the required interaction samples, and the training workload. It would also require temporal detail beyond what can be supported by interpolating hourly ERA5 winds and three-hourly HYCOM currents. Using such an interval would therefore create a false impression of higher temporal resolution without introducing genuinely new environmental information.

Each IBTrACS track is first linearly interpolated to the 10 min simulation interval. The cumulative along-track distance is computed using the local equirectangular approximation

$$\Delta s_k = \sqrt{\left[111\Delta\lambda_k \cos(\varphi_k)\right]^2 + \left(111\Delta\varphi_k\right)^2}, \qquad (8)$$

where $\Delta\lambda_k$ and $\Delta\varphi_k$ are the longitudinal and latitudinal increments in degrees. The constant 111 km is the mean meridional arc length per degree, and $\cos(\varphi_k)$ converts a longitudinal degree to the corresponding zonal distance at latitude φ_k. The target length of each segment is sampled uniformly from 300–500 km. The start of the next segment is shifted by a fixed 250 km along the track, producing an overlap of 50–250 km and an average overlap of approximately 150 km. A new segment is initialized only when at least 300 km of track remains; the final residual is discarded. Because segment endpoints are assigned to the nearest 10 min interpolation point, the realized segment lengths are 292.1–507.1 km (median 396.4 km).

The 500-TC archive yields 7,248 segments from 495 TCs with sufficient track length. The training split contains 450 TCs, of which 445 contribute at least one usable segment, yielding 6,480 training segments. The held-out test split contains 50 TCs and 768 segments. The training and test sets are separated at the TC level, so segments from the same TC never appear in both sets. Segment durations range from 2.2 to 103 h, with median and mean durations of 16.5 and 20.0 h, respectively.

Track segmentation is not merely a data-processing choice; it reflects the physical operating limit of the surface platform. The surface endurance of an SUV is finite, so one deployment cannot observe the complete life cycle of a TC. A 300–500 km along-track segment instead represents a self-contained mission in which the fleet can correct forecast-induced displacement, reach a physically relevant part of the TC, and collect a sustained observation sequence before the surface energy budget is exhausted. After entering the near-core region, the vehicles can dive and sample the upper ocean at several depths. The Blue Whale-class platform provides the required surface-transit and subsurface operating capabilities [28]. For the present mission concept, we assume a reconfigurable payload compatible with current profilers, CTD, and biogeochemical sensors, so the complete observing concept supports temperature, salinity, current velocity, nutrients, oxygen, and other biogeochemical variables. These measurements can characterize both the physical and biogeochemical responses of the upper ocean to TC forcing. The control experiments deliberately use current velocity only as the decision-making observation, avoiding model dependence on variables that are not consistently available across the complete 500-TC archive; this restriction does not imply that the platform is limited to current measurements in a real deployment.

4. Multi-Agent Reinforcement Learning Method

4.1. Blue Whale-Class Platform Modeling

The environment maintains each vehicle's horizontal position and propulsion velocity together with the local environmental field evaluated at that position. The surface actor observes the fleet-acquired 10 m wind patches and its own normalized propulsion velocity; the underwater actor observes a causal five-frame history of its local current speed and components together with its own normalized propulsion velocity. Accumulated path length and the propulsion-energy proxy are used for reward computation and evaluation rather than supplied as actor observations. Coastline distance, when enabled, enters safety filtering and the land-avoidance penalty and is also not a default actor input. Two surface maximum-speed scenarios are evaluated. The 12 m/s scenario is close to the current Blue Whale's public surface-design speed of 23 kn (≈11.83 m/s), whereas 20 m/s represents a potential future platform configuration with improved propulsion. For the subsurface stage, 1.5 m/s is close to the current Blue Whale's public submerged-design speed of 3 kn (≈1.54 m/s), whereas 2.0 m/s is a prospective future-upgrade assumption. State propagation follows the mode-dependent speed bound and horizontal motion model in Eqs. (1) and (2), with actions mapped to propulsion through Eq. (6). Surface motion neglects the direct current contribution because the 12–20 m/s propulsion speed is much larger than the

Table 2
Public Blue Whale reference and two simulation scenarios. The two simulation columns are modeling assumptions, not certified vehicle data.

Property	Public reference	Future-upgrade scenario	Current-platform scenario
Role	Engineering reference	Potential future capability	Near-current capability
Surface speed	23 kn, ≈11.83 m/s	20 m/s	12 m/s
Submerged speed	3 kn, ≈1.54 m/s	1.5 (near-current) or 2.0 (future) m/s	1.5 (near-current) or 2.0 (future) m/s
Submerged endurance	≥4 h at 3 kn; ≥72 h standby	20 h scenario	20 h scenario
Sampling depth	≈60 m	0–60 m (7-layer)	0–60 m (7-layer)
Payload	≥800 kg	800 kg	800 kg
Fleet size	Single platform	4/8/12/16	4/8/12/16
Decision interval	Not specified	10 min	10 min
Formation	–	100 km ring, $360°/N$	100 km ring, $360°/N$

typical surface-current contribution, and because the surface stage is intended to represent TC approach and high-wind-region tracking. In contrast, the subsurface model explicitly retains ambient advection because the 1.5–2.0 m/s propulsion speed is of the same order as the TC-induced current. The reduced point-mass formulation and its limitations are stated after Eq. (2). Table 2 separates the public engineering reference from the two simulation scenarios and identifies which values are modeling assumptions rather than certified platform specifications.

4.2. Decentralized Partially Observable Decision Process

The cooperative observation task uses the Dec-POMDP defined in Eq. (7).

For each vehicle i, the observation contains:

1. the normalized wind patches at 10 m above the sea surface sampled by all SUVs in the main surface experiments, or its own local current observation at the assigned ocean depth in the active underwater comparison;
2. its own velocity in the surface configuration;
3. a five-frame history of local current speed and components when the underwater active controller is used.

The final wind-only surface configuration is memoryless and assumes ideal communication, so every SUV receives the $3N$ wind-observation values acquired by the full fleet plus its own two normalized velocity components. By contrast, the underwater controller receives only its own local current and velocity channels; the shared policy parameters do not imply that observations, states, or actions are shared among vehicles. The main experiment does not provide the complete wind or current field, the global best-track center, or future reanalysis frames to the actor. The best track is used only to define the training target and to compute evaluation metrics.

The policy operates as a closed perception–decision–action loop rather than following a precomputed static route. At every 10 min decision interval, it observes the current environment, generates a new action, moves the vehicle, observes the updated environment, and repeats:

$$\begin{aligned}\mathbf{o}_{i,t} &\longrightarrow \mathbf{a}_{i,t} = \pi_\theta(\mathbf{o}_{i,t}) \\ &\longrightarrow \mathbf{x}_{i,t+1} \longrightarrow \mathbf{o}_{i,t+1}.\end{aligned} \tag{9}$$

This feedback process allows the fleet to revise its motion as the wind and current fields evolve. The observation is not only an output recorded after motion; it is also the next control input that determines subsequent motion. Here, "supervision" refers to this measurement-driven feedback role rather than to supervised learning with target action labels: the policy is trained from scalar rewards, but each new observation directly supervises the next decision through the closed-loop state. Because the coupled environment is nonstationary and stochastic, this observation–planning coupling is the principal distinction from static or decoupled trajectory-planning pipelines. The same mechanism supports adaptation to changing local observations, partial observability, and deployment uncertainty.

4.3. Surface and Subsurface Reward Design

The surface objective is to follow the translating TC as closely as possible and reach the high-wind region, with the final tracking error as the primary operational metric. Let d_i denote the distance from SUV i to the TC center. The surface reward used in the implementation is

$$\begin{aligned}r_i^{\mathrm{surf}} = &-\frac{\min(d_i, 300)}{100} + b_{\mathrm{near}}\mathbf{1}_{d_i<100} \\ &-\frac{W_{\mathrm{form}}\max(0, d_i - R_{\mathrm{form}})}{100} \\ &-\lambda_u\|\mathbf{a}_{i,t} - \mathbf{a}_{i,t-1}\|_2^2 + r_i^{\mathrm{coast}},\end{aligned} \tag{10}$$

where the first term is a bounded proximity cost and the second term provides a discrete near-center bonus with $b_{\mathrm{near}} = 1$. The third term adds an additional radius-excess penalty beyond $R_{\mathrm{form}} = 100$ km, and the fourth term penalizes abrupt action changes with $\lambda_u = 0.05$. The term r_i^{coast} represents the coastal-safety reward defined below. The 300 km cap prevents distant errors from producing an excessively steep gradient and reduces premature policy collapse during

early training. After aggregation over SUVs, the team average reward is clipped to $[-5, 2]$.

For subsurface observation, four SUVs are assigned to each prescribed depth. Their objective is not to follow the surface TC center directly, but to observe the highest possible current speed. Low-speed SUVs cannot generally overtake a translating TC, and the vertical projection of the surface center is not necessarily the location of the strongest subsurface current. For the active-control comparison, the current-seeking reward is

$$\begin{aligned} r_i^{\mathrm{sub}} = & \min\left(\frac{\|\mathbf{u}_{i,k}\|}{V_{\mathrm{norm}}}, R_{\max}\right) \\ & - \lambda_E \|\tilde{\mathbf{a}}_i\|_2^2 + \beta_{\parallel} \max(0, \cos\theta_i) \\ & + r_i^{\mathrm{coast}}, \end{aligned} \tag{11}$$

where $V_{\mathrm{norm}} = 1.0$ m/s, $R_{\max} = 2.0$, $\lambda_E = 0.05$, and $\beta_{\parallel} = 0.5$. The angle θ_i denotes the direction difference between propulsion and the pre-move current $\mathbf{u}_{i,k}$, so the positive-alignment term is activated only when the SUV pushes downstream; formally, $\cos\theta_i = (\tilde{\mathbf{a}}_i \cdot \mathbf{u}_{i,k})/(\|\tilde{\mathbf{a}}_i\|_2 \|\mathbf{u}_{i,k}\|_2)$ when the denominator is nonzero. The current-intensity reward is therefore evaluated at $\mathbf{x}_{i,k}$ before the state update. For the local maximum-current objective, passive drift is retained as the no-propulsion reference because it follows the ambient flow directly and does not depend on a learned reward policy.

4.4. Coastal Safety and Action Smoothing

The land mask is converted into a nonnegative offshore-distance field. A vehicle inside land receives a large penalty, and a vehicle within the buffer distance D_{buf} receives

$$r_i^{\mathrm{coast}} = \begin{cases} -P_{\mathrm{land}}, & D_i = 0, \\ -W_b \max(0, 1 - D_i/D_{\mathrm{buf}}), & 0 < D_i < D_{\mathrm{buf}}, \\ 0, & D_i \geq D_{\mathrm{buf}}, \end{cases} \tag{12}$$

where D_i is the offshore distance and P_{land} and W_b are positive penalty coefficients.

Three complementary mechanisms are used to reduce coastline intrusion: offshore-only rejection sampling during deployment, a soft nearshore penalty, and an action-level safety filter. The current main experiments use a minimum deployment distance of 100 km, a 100 km nearshore buffer, and a 100 km safety margin. The deployment step removes invalid initial states, the reward discourages trajectories near land, and the safety filter removes the onshore action component when an SUV approaches the shoreline. These mechanisms reduce land exposure but are not posed as an unconditional hard constraint on every vehicle position.

To reduce oscillatory control, the main surface experiments use an exponential action filter:

$$\mathbf{a}_t^{\mathrm{filt}} = (1 - \beta)\mathbf{a}_t^{\mathrm{raw}} + \beta \mathbf{a}_{t-1}^{\mathrm{filt}}, \tag{13}$$

where $\beta \in [0, 1)$ controls the degree of temporal smoothing. The filter uses only the current and previous actions and therefore does not introduce future-information leakage.

4.5. Parameter-Sharing Independent PPO

All SUVs contribute to the same observation and tracking objective, and the training return is defined as the team-average reward. The problem is therefore a fully cooperative task, rather than a competitive or mixed cooperative–competitive task. This cooperative reward structure motivates the use of IPPO, in which decentralized vehicle actions are optimized against a common team objective while policy parameters are shared across vehicles.

In this paper, IPPO denotes the cooperative and parameter-sharing implementation of independent PPO. All SUVs use the same actor and critic weights, while their observations and action samples remain vehicle-specific. Actions are sampled independently, and the per-agent values are summed for training without a global-state centralized critic. The main surface model uses a memoryless multilayer perceptron, whereas the active underwater controller uses a five-frame current history. The shared cooperative return is the team-average reward, and a single clipped PPO objective is optimized over the factorized multi-agent action vector:

$$\begin{aligned} L^{\mathrm{CLIP}}(\theta) = \mathbb{E}_t \Big[\min\Big(& \rho_t(\theta)\hat{A}_t, \\ & \mathrm{clip}\left(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon\right) \hat{A}_t \Big) \Big], \end{aligned} \tag{14}$$

where

$$\rho_t(\theta) = \prod_{i \in \mathcal{I}} \frac{\pi_\theta(a_{i,t} \mid o_{i,t})}{\pi_{\theta_{\mathrm{old}}}(a_{i,t} \mid o_{i,t})}, \tag{15}$$

and $\hat{A}_t$ is estimated by generalized advantage estimation from the team-average reward. The actor objective is complemented by a clipped value loss and an entropy bonus. During execution, each SUV computes its action from its own observation and the shared policy network.

Off-policy SAC and TD3 were not used for the final multi-SUV comparison. Both require a replay buffer, and a multi-agent transition contains the joint observation and action of up to 16 vehicles. The memory footprint therefore scales with replay capacity, fleet size, and the number of parallel environments. The final experiments prioritize stable large-scale parallel on-policy sampling and consequently use parameter-sharing IPPO. This choice is an algorithm-design constraint rather than a claim that IPPO is universally superior to SAC or TD3 in sample efficiency.

The complete reward coefficients, safety parameters, network dimensions, and PPO hyperparameters are summarized in Appendix A. The project repository and software package sources are listed in the Code and package availability section.

4.6. Surface Evaluation and Generalization

The center-distance reward is available only during training. It does not enter the actor observation and is absent during validation. Consequently, the surface results test a specific form of generalization: a frozen policy must infer where to move from shared wind observations in an unseen TC. The policy is not asked to read the best-track center,

and it cannot exploit a reward computed from that center. This distinction is essential for interpreting the 100–800 km curves as forecast-error tolerance rather than as supervised center tracking.

Once the high-wind region is reached, diving provides the second part of the sampling objective. The subsurface current there is expected to contain the upper-ocean response generated by the TC forcing and its preceding passage, including near-inertial and mesoscale variability. Sampling that dynamically conditioned region is expected to be more informative than descending in an unrelated background ocean. The present study evaluates the observation task itself; it does not quantify the subsequent impact of these measurements in a data-assimilation or forecast system. Operationally, surface observation benefit is therefore represented by the final tracking error (km), the fleet-averaged SUV-to-best-track-center distance at the final time step. A smaller value denotes a higher benefit because the fleet ends closer to the TC core. Time-averaged distance is retained as a secondary process diagnostic.

For training seed s and its validation dataset, the seed-level value is

$$D_{\mathrm{final}}^{(s)} = \frac{1}{M_s}\sum_{e=1}^{M_s}\frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{x}_{i,e}(T_e) - \mathbf{x}_{\mathrm{TC},e}(T_e)\right\|_2, \quad (16)$$

where T_e is the final decision time of validation segment e, and M_s is the number of validation segments. The reported final tracking error is the mean of $D_{\mathrm{final}}^{(s)}$ over the ten independent training seeds, with uncertainty expressed as the sample standard deviation across seeds.

The process-level fleet-mean time-averaged distance for the same seed is

$$D_{\mathrm{avg}}^{(s)} = \frac{1}{M_s}\sum_{e=1}^{M_s}\frac{1}{K_e+1}\sum_{k=0}^{K_e}\frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{x}_{i,e,k} - \mathbf{x}_{\mathrm{TC},e,k}\right\|_2, \quad (17)$$

where K_e is the final decision index of segment e, k indexes decision times from the initial step through T_e, and the other symbols follow Eq. (16). Thus, the distance is first averaged over the N SUVs at each time, then over the complete episode, then over the validation dataset. The reported fleet-mean time-averaged distance is the mean of $D_{\mathrm{avg}}^{(s)}$ over the ten training seeds, again with uncertainty reported as the sample standard deviation across seeds.

The final surface matrix uses only local wind observations and evaluates 4-, 8-, 12-, and 16-SUV formations. The vehicles in each formation have equal angular spacing of $360°/N$ on a 100 km ring. During training, the formation-center offset is sampled uniformly inside an 800 km disk around the initial TC center, representing the assumed upper bound of track-forecast position error. The trained policies are then evaluated without retraining at fixed formation-center offsets of 100, 200, 400, 600, and 800 km with random bearings.

For each fleet-size, deployment-radius, and speed setting, metrics were first averaged over the validation dataset within each training seed. Means and sample standard deviations were then computed across the ten independent seeds. Paired comparisons between 12 and 20 m/s used two-sided paired t-tests on the ten seed-level values. Because the twenty fleet-size/radius comparisons were tested separately without multiplicity correction, the resulting p-values are interpreted as exploratory evidence rather than confirmatory hypothesis tests.

4.7. Underwater Evaluation Protocol and Baselines

The final subsurface matrix assigns four SUVs to each ocean depth and uses 20 h episodes. Its objective is to observe the highest attainable local current speed at depths of 0, 10, 20, 30, 40, 50, and 60 m under maximum propulsion speeds of 1.5 and 2.0 m/s. It compares active IPPO, random control, and passive drift under otherwise identical conditions. Because the surface stage is responsible for approaching the TC and high-wind region, the subsurface rollout starts from the surface endpoint and initializes the four vehicles within 200 km of the TC center. The 100–800 km forecast-error protocol applies only to the surface stage.

No inter-vehicle communication is assumed below the surface. Each SUV conditions its action only on its own current speed and components, a causal five-frame current history, and its propulsion velocity. It receives neither the observations nor the positions or actions of the other SUVs. The policy parameters are shared, but information is not exchanged during execution. Increasing the fleet at one depth would enlarge the aggregate sampling footprint without providing additional information to an individual policy. The comparison therefore fixes the fleet at four vehicles per depth and pairs all methods on the same episodes and deployment seeds. In a larger deployment, four vehicles can be assigned to each of several depths.

The three strategies differ only in how propulsion is generated:

$$\begin{aligned}\mathbf{a}_i^{\mathrm{active}} &= \mathrm{clip}\left(\pi_\theta(\mathbf{o}_i), -1, 1\right),\\ \mathbf{a}_i^{\mathrm{random}} &\sim \mathcal{U}\left([-1,1]^2\right),\\ \mathbf{a}_i^{\mathrm{drift}} &= \mathbf{0}.\end{aligned} \quad (18)$$

During evaluation, the active IPPO policy uses its deterministic mean action, whereas random control resamples a two-dimensional action every 10 min. Active IPPO and random control share the same propulsion limit and equations of motion; only the active policy conditions its action on the local current history. Passive drift applies zero propulsion and advances solely by ambient current advection.

For vehicle i at depth z, the ground-path length after K decision steps and the mean experienced current speed are

$$L_i(z) = \sum_{k=0}^{K-1}\left\|\mathbf{x}_{i,k+1} - \mathbf{x}_{i,k}\right\|_2, \quad (19)$$

$$\bar{V}_i(z) = \frac{1}{K+1}\sum_{k=0}^{K}\left\|\mathbf{u}_{\mathrm{ocean}}(\mathbf{x}_{i,k}, z, t_k)\right\|_2. \quad (20)$$

$$C(z) = A_{\text{cell}} \left| \bigcup_{i=1}^{N} \bigcup_{k=0}^{K} \mathcal{G}(\mathbf{x}_{i,k}) \right|, \tag{21}$$

where $\mathcal{G}(\mathbf{x})$ is the 0.05° grid cell containing position $\mathbf{x}$, $|\cdot|$ counts unique cells, and A_{cell} is the cell area in km^2. The figures report the fleet means $N^{-1}\sum_{i=1}^{N} L_i(z)$ and $N^{-1}\sum_{i=1}^{N} \bar{V}_i(z)$, together with the propulsion-energy proxy $N^{-1}\sum_{i=1}^{N}\sum_k \|\tilde{\mathbf{a}}_{i,k}\|_2^2$.

5. Simulation and Results

5.1. Surface Active Tracking

Using the surface protocol in Section 4.6, we evaluate the extent to which the trained policies reduce forecast-induced deployment errors and approach the moving TC center and high-wind region.

Figure 3 reports the primary final tracking error from Eq. (16), for which smaller values are better, and the secondary time-averaged distance. Each value summarizes the validation dataset over ten independent training seeds.

Using the final tracking error as the operational observation benefit, the right panel of Fig. 3 reports the primary speed-limit comparison. Averaged over the four fleet sizes and ten training seeds, the 20 m/s policy obtains final tracking errors of 197.1, 209.2, 260.9, 362.1, and 524.0 km at deployment radii of 100, 200, 400, 600, and 800 km, respectively. The corresponding 12 m/s values are 221.0, 238.9, 317.5, 450.9, and 635.1 km. Thus, 20 m/s yields a lower final tracking error in all 20 matched fleet-size/radius comparisons, corresponding to a higher operational observation benefit. Its fleet-averaged advantage increases from 23.8 km at 100 km to 111.1 km at 800 km.

The absolute distances still increase with deployment radius for both speed limits, which is expected because a finite propulsion speed cannot eliminate a larger initial displacement within every segment duration. The objective is therefore to minimize the final tracking error at each error level rather than to make it independent of deployment error. The final-tracking-error advantage of 20 m/s is larger than the corresponding time-averaged advantage: 23.8 versus 10.1 km at 100 km and 111.1 versus 84.2 km at 800 km. This indicates that the higher-speed policy continues correcting residual separation late in the episode rather than merely remaining closer during the initial transit.

This benefit involves a clear propulsion trade-off. Relative to 20 m/s, the 12 m/s policy reduces path length per SUV by 78.7, 95.5, 147.4, 196.6, and 233.3 km at the five radii and lowers mean ground speed by 1.74–4.18 m/s. Accordingly, 20 m/s provides the higher final observation benefit, whereas 12 m/s is preferable when shorter trajectories and lower propulsion use are prioritized.

Time-averaged distance provides a secondary process-level diagnostic of whether proximity is maintained throughout an episode. Across the twenty matched fleet-size/radius comparisons, the 20 m/s policy has the smaller mean time-averaged distance in all 20 cases. The median paired difference ($d_{12} - d_{20}$) is +51.8 km. Uncorrected two-sided paired tests indicate lower distance at 20 m/s in 19 cases and at 12 m/s in none of the cases at $p < 0.05$. Because these comparisons were not adjusted for multiplicity, they provide exploratory rather than confirmatory evidence.

The relative time-averaged distance penalty of the 12 m/s policy is +6.3% at 100 km, +10.6% at 200 km, +18.2% at 400 km, +17.7% at 600 km, and +14.0% at 800 km. The non-monotonic relative pattern, together with the monotonic increase in absolute final-tracking-error benefit, is consistent with the finite time available to remove forecast-induced displacement. Overall, the results show that the policy learned from wind observations alone provides increasing operational value for large initial displacement errors when a 20 m/s speed envelope is available, while the 12 m/s envelope offers a lower-effort alternative.

Taken together, Figs. 3–4 and Table 3 connect the operational benefit metric to both population-level statistics and representative trajectories. The right panel of Fig. 3 is the primary benefit comparison: for each deployment radius, the preferred configuration is the one with the smallest final tracking error. Figure 2 shows the mechanism in the four-SUV, 400 km case, where the formation first turns along the observed wind structure and then reduces its separation from the TC center. Figure 4 shows that this inward correction persists from 100 to 800 km, although the final marker remains farther from the center as the initial deployment error increases. These representative paths explain why an effective controller can reduce error without making the final tracking error independent of deployment radius.

Table 3 provides a same-policy check of the benefit interpretation. Applying the execution filter lowers the fleet-mean final tracking error from 346.4 to 292.6 km, a 53.8 km improvement, while also shortening trajectories and reducing mean and peak speed. Thus the lower final tracking error, and therefore the higher operational observation benefit, is not obtained by simply commanding more aggressive propulsion. The trajectory figures and the ablation table jointly indicate that better final placement results from smoother closed-loop correction of the observed wind pattern.

To provide an internally consistent visual comparison, Figs. 2–5 use the same held-out CHAMPI (2015) segment. The surface figures show motion from the segment start to its endpoint, while the subsurface rollout starts from the segment-end anchor, representing the sequential transition from surface delivery to underwater observation.

Figure 2 resolves the temporal evolution for the four-SUV, 400 km case. At deployment, the four vehicles are dispersed around the formation center far from the TC. During the first several hours, the vehicles turn toward the strengthening wind region and begin to reduce the initial displacement. By 6.5 h, all four vehicles have initiated the inward turn; by 13.0 h they have entered the outer high-wind region; and by 19.5 h all four trajectories have turned into

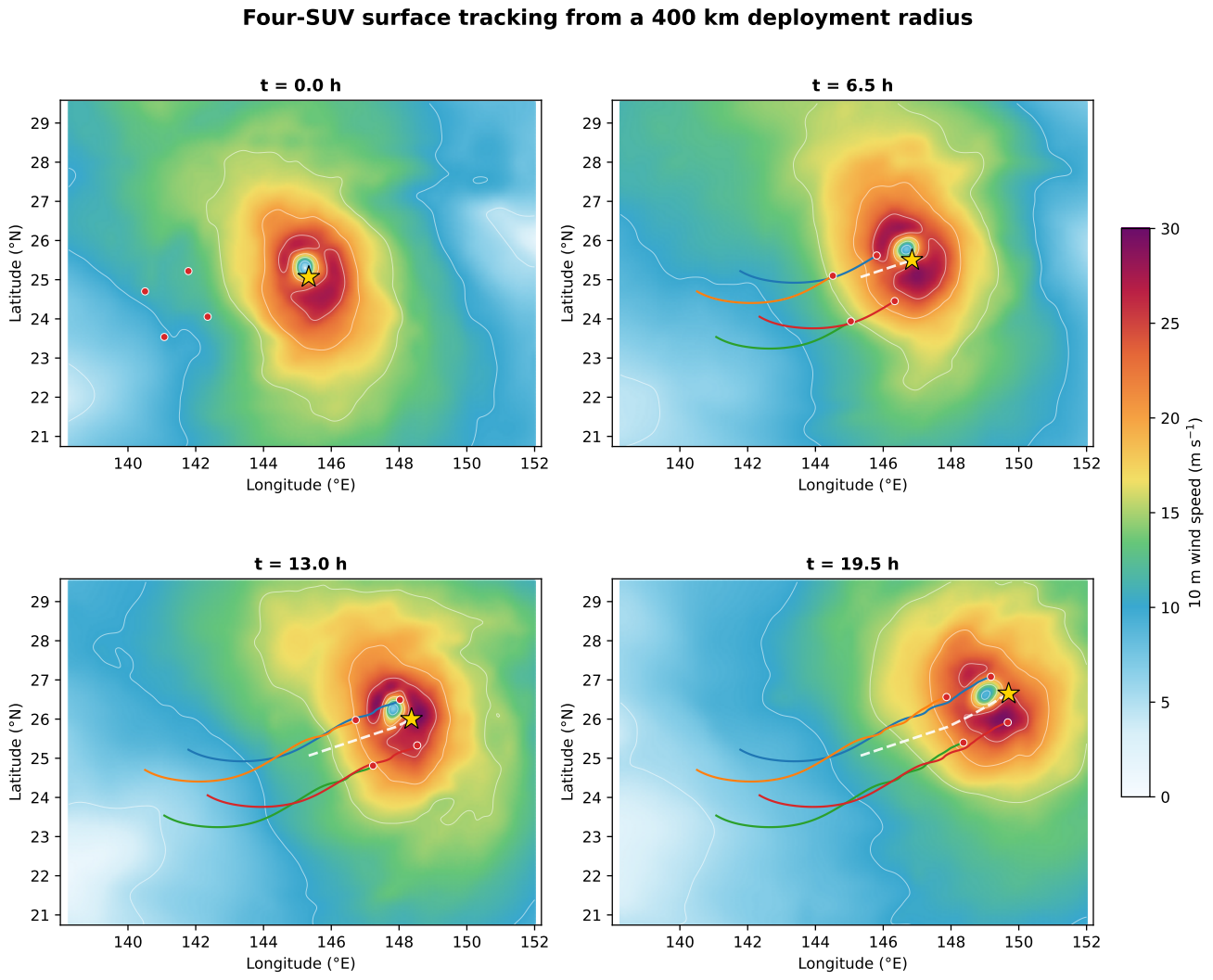


Figure 2: Four-SUV surface-tracking process for CHAMPI (2015) at the 20 m/s speed limit. The four panels show the initial deployment and subsequent states at approximately 6.5, 13.0, and 19.5 h. The color background is the spatially interpolated ERA5 10 m wind-speed field at each time. Small red dots mark vehicle positions, and the yellow star marks the TC center.

the high-wind branch and substantially reduced their initial displacement. The individual paths differ because each vehicle has its own state and position within the shared fleet-level wind observation; nevertheless, the fleet-level motion follows the wind pattern toward the TC core and illustrates long-range error correction in this representative episode.

Figure 4 shows how initial forecast error changes the surface search process. At 100–200 km, the formations converge rapidly and remain concentrated around the TC core. As the deployment radius increases to 600–800 km, the trajectories lengthen and fan out over a wider region before turning toward the TC. Increasing the fleet from 4 to 16 SUVs produces a denser set of sampled branches and broader spatial coverage, but it does not remove the need for long-range transit. In the 12- and 16-SUV cases, many paths overlap near the core because the shared wind observations guide the vehicles toward the same high-wind region. In these representative cases, fleet size mainly broadens the sampled approach region, whereas deployment error primarily determines the required transit distance.

The 600 and 800 km rows also reveal two coupled failure mechanisms at large initial displacement. At 600 km, the representative fleet can still recover a usable inward direction from the observed wind-speed field and its spatial gradient, so the vehicles continue toward the TC despite the longer transit. At 800 km, the vehicles must first overcome a larger geometric separation with finite speed and a finite episode duration; in addition, the TC-induced wind speed and its local gradient become weaker and less distinctive at that range. The reduced signal quality makes the approximate bearing toward the TC more ambiguous, so the policy may approach more slowly, change direction repeatedly, or fail to recover the TC within the episode. These two effects are coupled: longer travel time increases the consequence of even a small directional error, while the weaker remote wind signature makes such an error more likely.

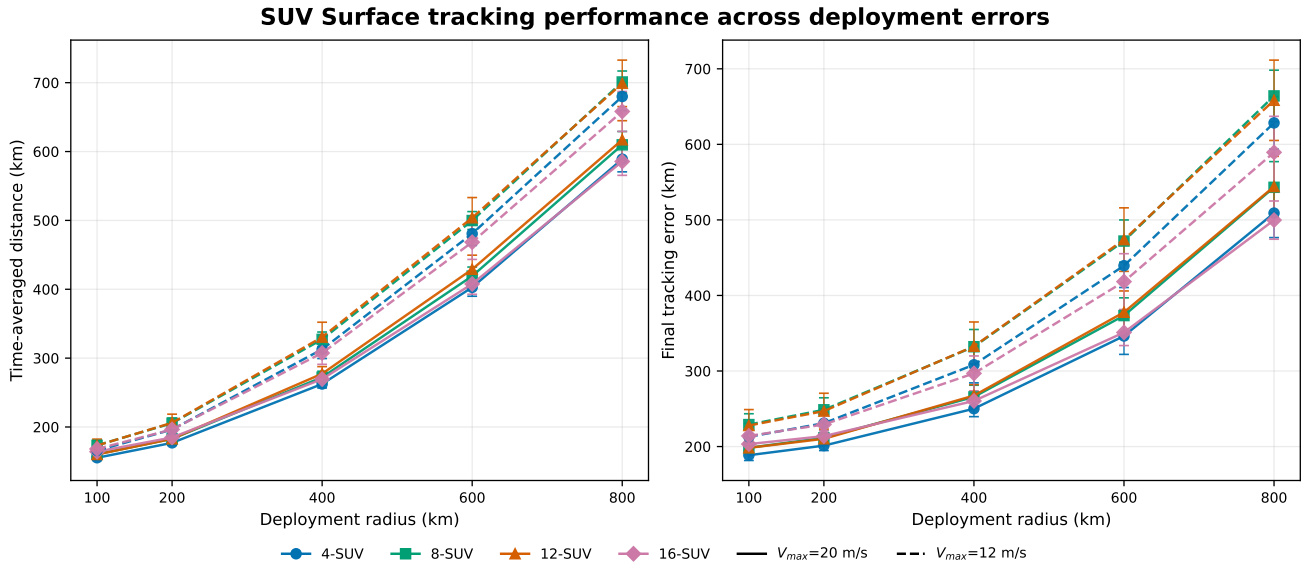


Figure 3: SUV surface tracking performance across deployment errors. The left panel shows the fleet-mean time-averaged distance to the TC center, representing whether the fleet remains close to the moving TC throughout an episode. The right panel shows the final tracking error to the TC center at the end of the episode; smaller values indicate higher operational observation benefit. Colors denote fleet sizes and line styles denote the 20 and 12 m/s speed limits. Each point summarizes ten independent training runs (ten random seeds), with error bars showing the sample standard deviation.

5.2. Execution-Layer Action Filtering

Table 3 reports the low-pass filtering ablation over the same trained actor, validation dataset, and 10 min decision interval. Without filtering, the 4-, 8-, 12-, and 16-SUV configurations travel 1294.9–1331.5 km per vehicle, with mean speeds of 19.23–19.73 m/s and maximum speeds of 19.92–20.00 m/s. With the execution-layer filter ($\beta = 0.8$), path length falls to 690.8–769.8 km, mean speed to 11.11–12.72 m/s, and maximum speed to 16.13–18.20 m/s. The final tracking error changes from 288.2 to 258.2 km for four SUVs, from 305.2 to 308.6 km for eight SUVs, from 478.5 to 330.0 km for 12 SUVs, and from 313.6 to 273.4 km for 16 SUVs. Averaged over the four fleet sizes, it decreases from 346.4 to 292.6 km, a reduction of 53.8 km (15.5%). Thus, filtering not only avoids degrading final tracking but improves it on average: three fleet sizes improve, whereas the eight-SUV case increases only slightly, by 3.4 km.

The same comparison shows reductions for every fleet size: path length decreases by 563.4, 547.1, 618.5, and 573.6 km for 4, 8, 12, and 16 SUVs, respectively, while maximum speed decreases by 1.80, 2.29, 3.87, and 2.94 m/s. The 12-SUV configuration has the largest relative path-length reduction, although the effect is not monotonic in fleet size. These results show that the execution filter improves final tracking on average while substantially shortening trajectories and reducing mean and peak speed.

5.3. Speed-Limit and Deployment-Error Analysis

The 12 and 20 m/s cases answer different engineering questions. The 12 m/s value is close to the published surface-speed scale of the current Blue Whale, whereas 20 m/s represents a potential future high-speed platform configuration rather than a certified current capability. A simple direct-pursuit estimate illustrates why deployment radius changes the effect of this choice. If the TC translates at speed U_c and the SUV moves toward it at V_{max}, the nominal closure rate is approximately

$$V_{close} = V_{max} - U_c. \tag{22}$$

For a deployment error D, the corresponding correction time is

$$t_{corr} \approx \frac{D}{V_{close}}. \tag{23}$$

This is only a kinematic lower bound because it neglects turning and imperfect center estimation, but it exposes the dominant scale. At 100–200 km, both speed limits provide enough closure to correct a moderate forecast displacement. At 600–800 km, a TC translation speed of 4–7 m/s removes a substantial fraction of the 12 m/s speed budget. The required correction time can then approach or exceed the 20 h mean segment duration. A 20 m/s vehicle retains a larger reserve for reducing initial center error, but its benefit is expected mainly at the larger radii; at short radii both cases already have sufficient speed.

The growing 12/20 m/s final-tracking-error gap follows this closure-rate scaling: both speeds are adequate at 100–200 km, whereas the available reserve of the 12 m/s platform contracts at 400–800 km. The 800 km test is therefore conservative for the lower-speed platform. Operationally, large forecast errors favor a faster vehicle, a tighter forecast-error envelope, or earlier deployment; 12 m/s remains preferable only when shorter paths and lower propulsion use are prioritized.

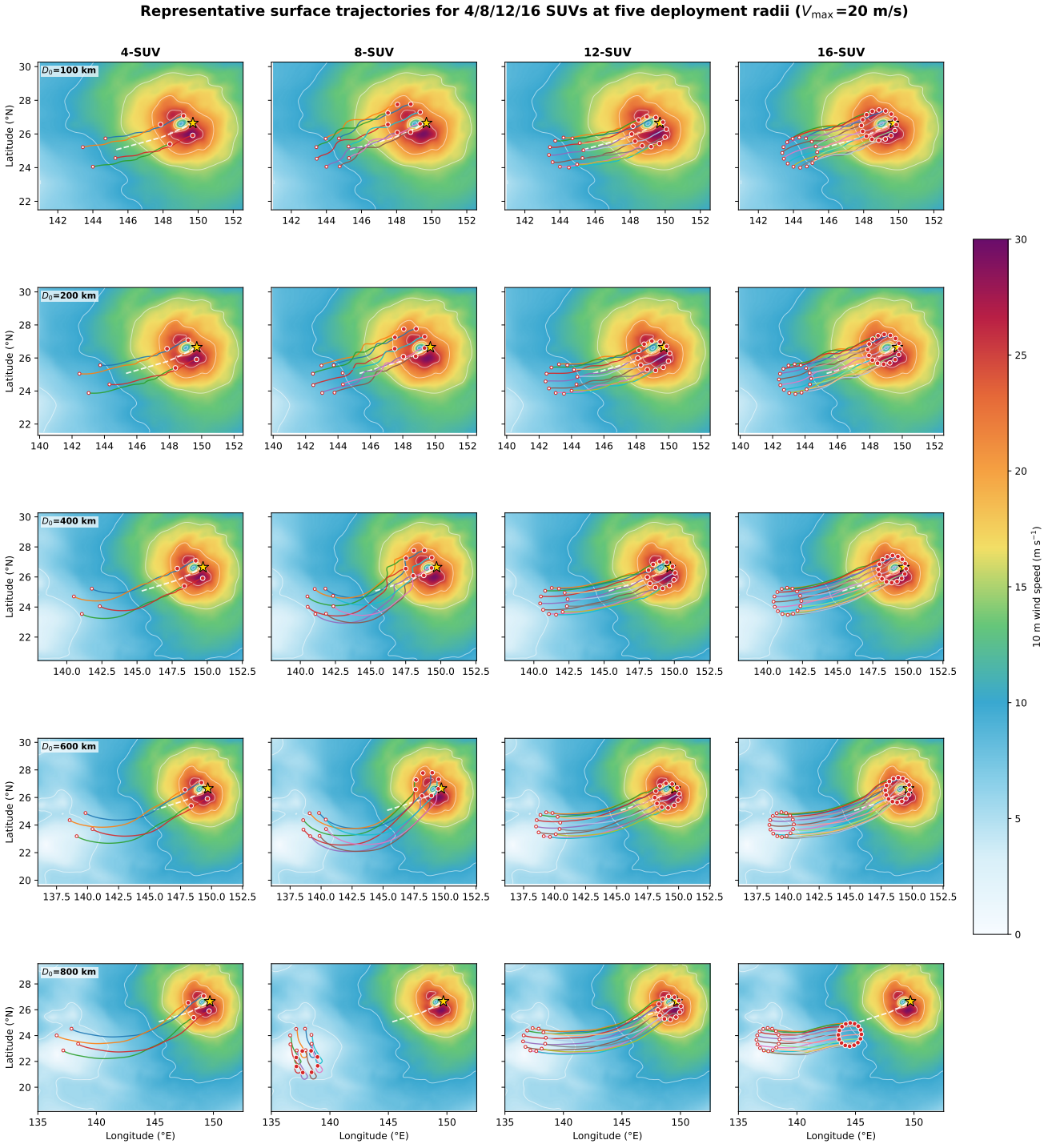


Figure 4: Representative CHAMPI (2015) trajectories for the final 20 m/s surface policy. Columns show 4-, 8-, 12-, and 16-SUV formations, while rows show nominal formation-center deployment radii of 100, 200, 400, 600, and 800 km. The color background is the spatially interpolated final-time ERA5 10 m wind-speed field. Thin colored lines are vehicle trajectories, open circles mark initial positions, small red dots mark final vehicle positions, and the yellow star marks the TC center at the end of the episode.

5.4. Underwater Strategy Comparison

For each depth, four SUVs seek higher current speed; the central comparison is therefore the fleet-mean current speed experienced along each trajectory, together with its path and propulsion cost. The comparison follows the protocol in Section 4.7. Figure 5 first illustrates the resulting trajectory structure for the three strategies.

Figure 5 illustrates the qualitative trajectory construction for the three strategies at shallow, intermediate, and deep sampling layers. Across these three representative depths,

Table 3
Effect of execution-layer low-pass filtering for 4/8/12/16 SUVs on the 500-TC benchmark. The same trained policy is evaluated with the filter off and on ($\beta = 0.8$) on the validation dataset.

Fleet	Filter	Final tracking error (km)	Path per SUV (km)	Mean speed (m/s)	Maximum speed (m/s)
4	Off	288.2	1331.5	19.73	20.00
4	On	**258.2**	**768.1**	**12.72**	**18.20**
8	Off	**305.2**	1318.0	19.57	19.97
8	On	308.6	**770.9**	**12.58**	**17.68**
12	Off	478.5	1309.3	19.34	20.00
12	On	**330.0**	**690.8**	**11.11**	**16.13**
16	Off	313.6	1295.0	19.23	19.92
16	On	**273.4**	**721.4**	**11.94**	**16.98**

Final tracking error is the fleet average of the individual SUV-to-best-track-center distances at the final time step. Off/on changes only the execution-layer filter on the same trained actor.

the mean experienced current speeds of active control, random control, and passive drift remain close within each depth, whereas their path lengths differ substantially: active control requires the longest paths and passive drift the shortest. The background fields also show that the spatial current structure changes with depth, which is why a single-depth example is insufficient. These episodes remain illustrative; all comparative inference is deferred to the final ten-seed, seven-depth evaluation.

Figure 6 reports the final ten-seed results at all seven depths from 0 to 60 m. At every depth and both propulsion limits, active IPPO attains a higher mean experienced current speed than passive drift and random control. The improvement over passive drift ranges from 2.1% to 5.4%, while the corresponding improvement over random control ranges from 2.0% to 5.1%. Random control remains close to passive drift in mean current speed but travels substantially farther.

The longer trajectories of active IPPO are an intended observation advantage, not merely a motion cost. Its path is 3.6–7.3 times that of passive drift, allowing the four SUVs to move through a larger spatial footprint and acquire more current profiles and contextual samples. The corresponding coverage increases by roughly 3.3–5.7 times. Across the 140 seed-configuration comparisons, the active policy has a higher mean experienced current speed in 133 cases; uncorrected paired tests classify 117 as significant improvements and three as significant deficits. Therefore, active control is not judged only by a small increase in mean current speed: it also increases the amount and spatial extent of the collected underwater data. Passive drift remains the low-motion reference and cannot deliberately relocate toward higher-current regions.

The propulsion limit consistently affects trajectory extent. At all seven depths, the 2.0 m/s policy travels 1.24–1.31 times farther than the 1.5 m/s policy because the larger speed envelope permits more movement within the same 20 h episode. This result links future platform speed directly to observation capacity: a faster SUV can cover more of the TC-induced current field and collect more profiles, although the associated energy and control constraints must still be considered. Mean experienced drift current decreases from 0.521 to 0.292 m/s between 0 and 60 m, and passive-drift paths contract from 33.6 to 17.1 km. Active paths decrease more moderately, from 122.1 to 98.7 km at 1.5 m/s and from 155.1 to 124.6 km at 2.0 m/s. The increasing active/passive path ratio with depth therefore reflects both weaker ambient advection and the largely depth-independent propulsion envelope, rather than an increasing policy advantage.

This behavior is consistent with the speed scales in the near-core region. For the paired 20 h episodes initialized within 200 km of the TC center, the area-averaged current speed at the evaluated 0, 30, and 50 m depths is 0.34–0.57 m/s, while the spatial 95th percentiles are 0.92–1.30 m/s and the 99th percentiles are 1.30–1.85 m/s. Strong local-current maxima reach 2.19–3.55 m/s across these depths, and the average local maximum within an instantaneous 200 km core area is 0.91–1.24 m/s. Thus, the high-current tail and local maxima are comparable to, or even exceed, the SUV propulsion limits of 1.5 and 2.0 m/s. These scales allow active propulsion to relocate a vehicle toward stronger currents, whereas passive drift follows the ambient flow at zero propulsion effort. The completed comparisons show that active control uses this capability to obtain both a modestly higher mean current speed and a substantially larger observation footprint and data volume.

6. Conclusion

This study introduced a simulation-based parameter-sharing IPPO framework for coordinated surface–subsurface tropical-cyclone observation with Blue Whale-class SUVs. The 12 m/s surface and 1.5 m/s submerged limits approximate the current Blue Whale design speeds, whereas 20 m/s and 2.0 m/s represent possible future upgrades; these near-current and future-upgrade settings were compared within their respective tasks. Event-specific environments aligned ERA5 10 m winds, HYCOM currents at seven depths from 0 to 60 m, and IBTrACS tracks for 500 TCs, with 450 TCs used for training and 50 unseen TCs for testing.

The surface and subsurface tasks have different objectives. At the surface, the fleet uses shared wind observations

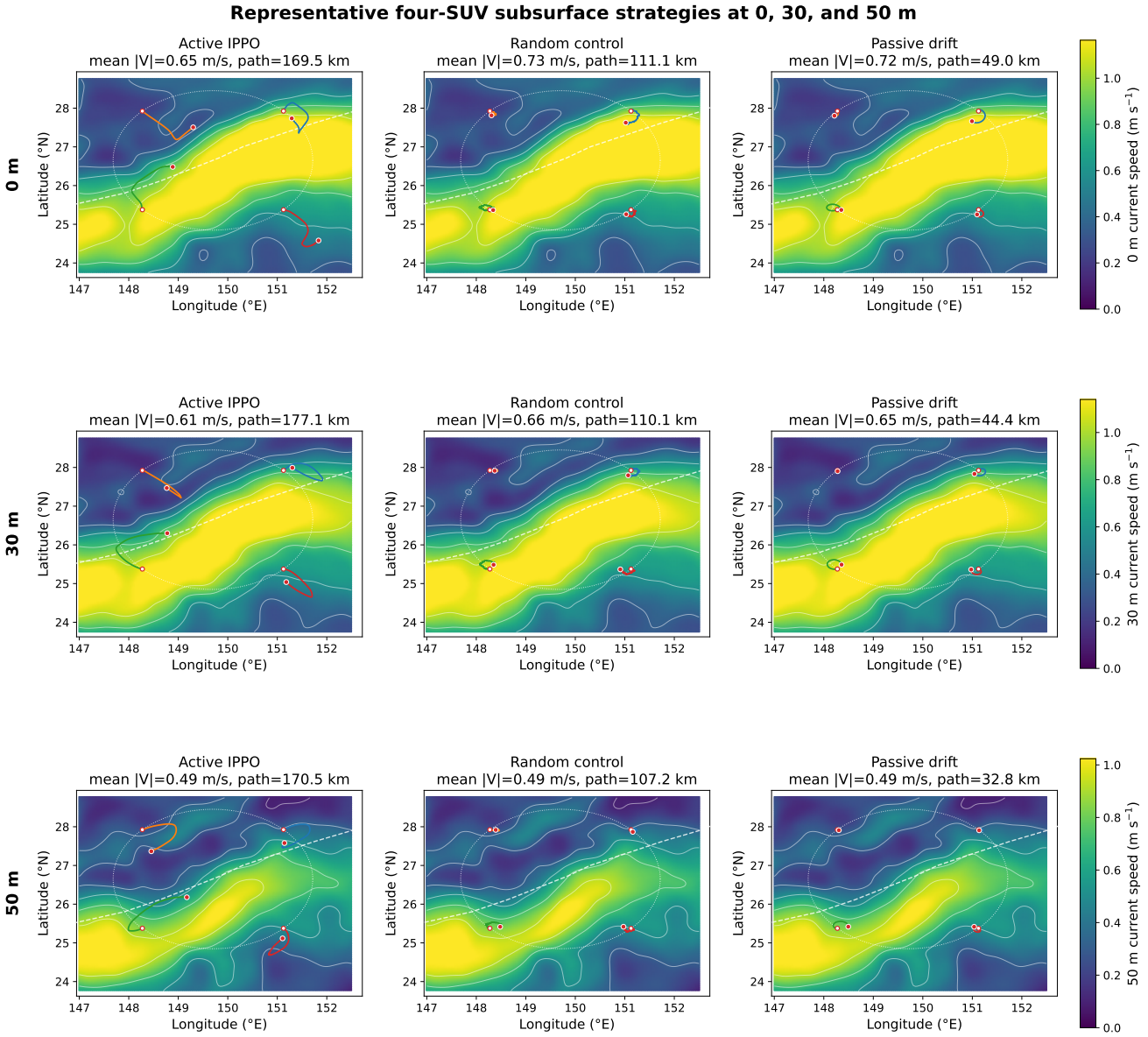


Figure 5: Representative four-SUV trajectories at 0, 30, and 50 m for the same CHAMPI (2015) segment used in the surface trajectory figures. These three depths provide shallow, intermediate, and deep examples from the full 0–60 m evaluation matrix. Rows denote depth and columns denote Active IPPO, random control, and passive drift. The propulsion limit is 2.0 m/s and the episode duration is 20 h. Background color is the spatially interpolated episode-mean HYCOM current speed at the corresponding depth; dashed white curves show the TC best track, open circles mark the common 200 km initial ring, and small red dots mark final vehicle positions.

to follow the TC and reach its high-wind region; the final tracking error measures how close the randomly deployed SUVs finish to the TC center. Across ten seeds, the 20 m/s policy achieved smaller time-averaged tracking distances than 12 m/s, and its advantage increased with deployment error. The 12 m/s policy instead produced shorter paths and lower speeds. Low-pass filtering reduced the fleet-mean final tracking error from 346.4 to 292.6 km while also shortening trajectories and reducing peak speed. These results show that the closed observation–decision–motion loop can guide the fleet inward using wind observations alone during validation.

At depth, four SUVs are assigned to each layer and seek the strongest observable current rather than the surface-center projection. Active IPPO, random control, and passive drift are compared under identical episodes and propulsion limits. Across all seven depths, active IPPO raises the mean experienced current speed by 2.1–5.4% and produces trajectories 3.6–7.3 times longer than passive drift, together with about 3.3–5.7 times greater spatial coverage. The learned propulsion therefore serves two purposes: it guides the SUVs toward higher-current water and keeps them moving over a wider area, producing more profiles and more contextual observations than a drift-only strategy.

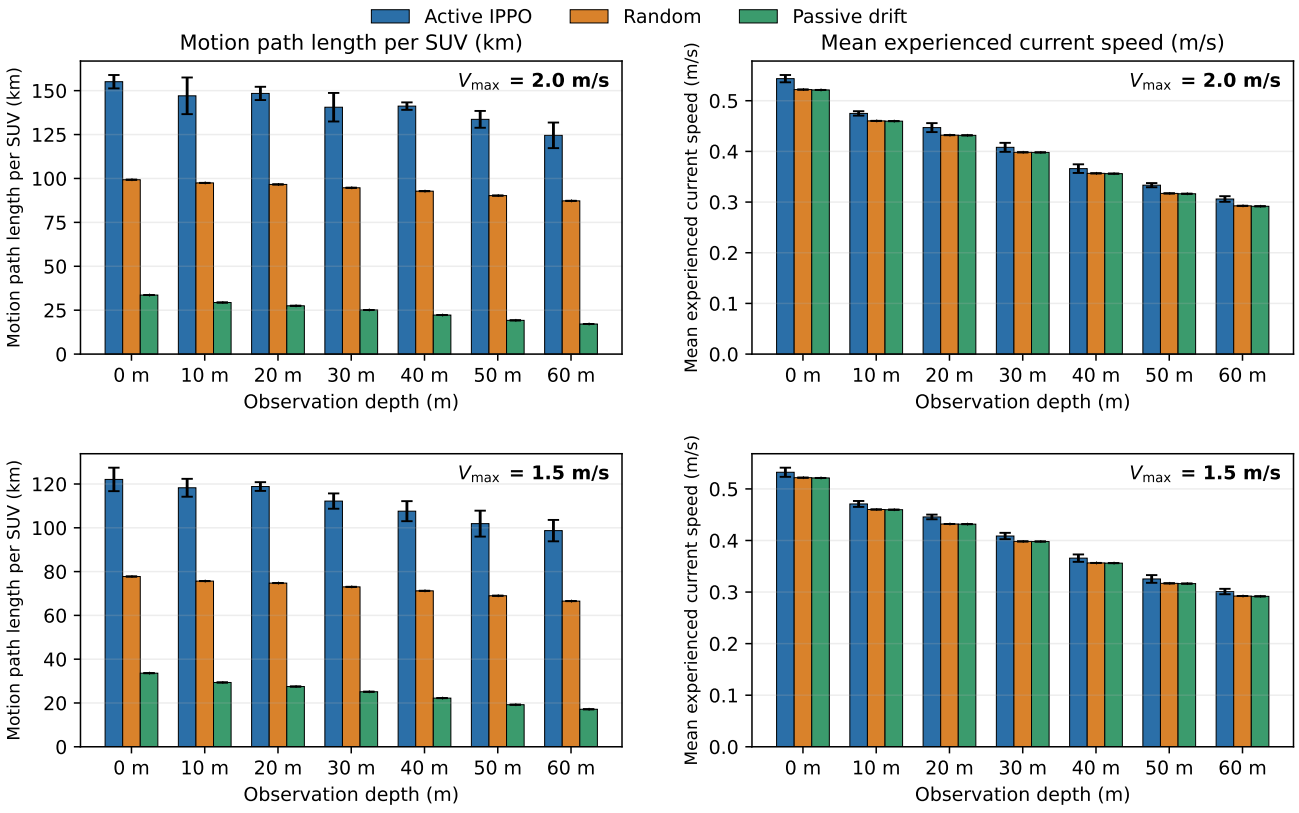


Figure 6: Comparison of active control, random control, and passive drift in the 20 h subsurface task at seven depths from 0 to 60 m. Rows show maximum propulsion speeds of 2.0 and 1.5 m/s; columns report the fleet-mean motion path length and mean experienced current speed. Error bars show the sample standard deviation across ten active-policy training seeds. Random control and passive drift are paired baselines without training-seed variability.

The 2.0 m/s policy travels 1.24–1.31 times farther than the 1.5 m/s policy at every depth. A larger speed envelope permits more movement within the same 20 h mission and therefore creates more opportunities to observe TC-induced currents. This identifies a direct path for future platform development: increasing SUV speed could raise the observation benefit for subsurface temperature, salinity, current, and biogeochemical fields, provided that energy use and control stability remain acceptable. The full seven-depth matrix therefore supports active sampling as an observation-capacity strategy while also exposing the propulsion-cost trade-off that must be managed operationally.

The present evidence is simulation-based and uses a mission-level horizontal point-mass model. Field validation, operational forecast impact, and calibrated six-degree-of-freedom vehicle dynamics remain future work.

Code and package availability

The code and processed results are available at `https://github.com/DYHAI/SUV-MARL-TC`. Experiments used Python 3.11; Stable-Baselines3 2.9.0; Gymnasium 1.2.2; PyTorch 2.11.0+cu128; NumPy 2.4.6; pandas 2.3.3; xarray 2026.7.0; SciPy 1.17.1; Matplotlib 3.11.1; and Ray 2.57.0 on a 24-core workstation with 64 GB DDR5 memory and an NVIDIA RTX 5060 Ti 16 GB GPU.

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A. Reward and IPPO Hyperparameters

Table A.1 summarizes the reward, action-filter, and safety parameters used in the surface and underwater experiments. Table A.2 lists the shared IPPO network and training hyperparameters.

Table A.1: Reward, action-filter, and safety parameters used in the final surface and underwater experiments.

Parameter	Symbol	Value	Description and applicable task
Panel A: surface reward and controller parameters			
Proximity-cost distance cap	$D_{\max}$	300 km	Distance cost is capped at 300 km to prevent excessively large gradients from distant initial deployments.
Near-center threshold	d_{near}	100 km	Threshold defining the discrete near-center bonus.
Near-center bonus	b_{near}	1.0	Bonus assigned when an SUV is within 100 km of the TC center.
Radius-excess threshold	R_{form}	100 km	Reference radius used by the additional distance penalty beyond 100 km.
Radius-excess weight	W_{form}	1.0	Weight of the penalty applied when an individual SUV is farther than 100 km from the TC center.
Action-change weight	λ_u	0.05	Weight of the squared action change $\|\mathbf{a}_{i,t} - \mathbf{a}_{i,t-1}\|_2^2$ at the 10 min interval.
Action-filter coefficient	β_{filt}	0.8	Coefficient of $\mathbf{a}_t^{\text{filt}} = (1-\beta)\mathbf{a}_t^{\text{raw}} + \beta\mathbf{a}_{t-1}^{\text{filt}}$. This is a controller setting rather than a reward term.
Surface reward clipping range	$[r_{\min}, r_{\max}]$	$[-5, 2]$	Clipping range applied to the team-averaged surface reward.
Panel B: underwater active-control comparison parameters			
Current-speed normalization	V_{norm}	1.0 m/s	Current-intensity reward is implemented as $\min(\|\mathbf{u}\|/V_{\text{norm}}, R_{\max})$.
Current-reward cap	$R_{\max}$	2.0	Upper bound of the normalized current-speed reward.
Propulsion-energy weight	λ_E	0.05	Weight of the propulsion penalty $\lambda_E\|\tilde{\mathbf{a}}_i\|_2^2$ in the current-seeking reward.
Downstream-alignment weight	$\beta_{\parallel}$	0.5	Weight of the positive propulsion–current alignment term.
Underwater reward clipping range	$[r_{\min}, r_{\max}]$	$[-5, 5]$	Clipping range applied to the team-averaged underwater reward during active control training.
Panel C: shared coastal-safety parameters			
Land penalty	P_{land}	20	Penalty applied when an SUV enters a land cell. Shared by the surface and underwater environments.
Nearshore buffer distance	D_{buf}	100 km	Width of the soft nearshore safety buffer.
Nearshore penalty weight	W_b	8.0	Maximum soft-penalty weight within the 100 km nearshore buffer.
Minimum deployment offshore distance	D_{deploy}	100 km	Minimum coastline distance required by offshore-only rejection sampling during deployment.
Action-filter safety margin	D_{filter}	100 km	Distance below which the onshore component of the commanded action is removed.

Note: the final subsurface comparison consists of active IPPO, random control, and passive drift. Passive drift does not optimize a learned propulsion policy or reward.

Table A.2: Parameter-sharing IPPO network and training hyperparameters used in the final surface and subsurface comparisons.

Parameter	Symbol	Value	Description
Algorithm	–	PPO	Stable-Baselines3 PPO with parameter sharing and independent per-agent action sampling.
Return aggregation	–	Team average	The environment reward is averaged across the N SUVs before computing the shared PPO return.
Optimizer	–	Adam	Adam optimizer used for the actor and critic parameters.
Policy network	–	2×256 MLP	Actor: Linear(d_{obs}, 256)–Tanh– Linear(256, 256)–Tanh– Linear(256, d_{act}).
Value network	–	2×256 MLP	Critic: Linear(d_{obs}, 256)–Tanh– Linear(256, 256)–Tanh– Linear(256, 1). Per-agent values are summed during training.
Per-agent observation dimension	d_{obs}	14/26/38/50	Surface wind-only observations for 4/8/12/16 SUVs, respectively. The four-SUV underwater active-control comparison uses $d_{\text{obs}} = 17$.
Surface observation	–	Wind at 10 m above sea level	Using ideal surface communication, each SUV receives the current wind speed and components at 10 m above the sea surface observed by every SUV in the fleet.
Underwater observation	–	Current only	With no inter-vehicle communication, the active IPPO controller observes only its own local current speed and components at the assigned ocean depth, together with a five-frame history and its own normalized propulsion velocity.
Action dimension	d_{act}	2 per SUV	Continuous east–north velocity command in $[-1, 1]^2$, followed by circular projection to enforce the maximum-speed constraint.
Parallel environments	N_{env}	4	Number of vectorized training environments.
Rollout length	n_{steps}	2048	Number of steps collected per environment before each PPO update.
Mini-batch size	B	256	Mini-batch size used during each gradient update.
Optimization epochs	N_{epochs}	10	Number of passes over each rollout.
Learning rate	α	3×10^{-4}	Adam learning rate used for the actor and critic.
Discount factor	γ	0.99	Reward discount factor.
GAE parameter	λ_{GAE}	0.95	Generalized advantage estimation applied to the team-average reward.
Policy clipping range	ϵ	0.2	Clipping range of the PPO probability ratio.
Value clipping range	ϵ_v	0.2	Clipping range of the value-function update.
Entropy coefficient	c_{ent}	0.01	Entropy regularization coefficient.
Value-loss coefficient	c_v	0.5	Weight of the value-function loss.
Maximum gradient norm	$g_{\max}$	0.5	Global gradient clipping threshold.
Training steps	T	2×10^6	Nominal environment steps per training run.
Independent seeds	–	0–9	Ten random seeds for each selected surface or active-control configuration.
Surface training forecast-error radius	R_{train}	0–800 km	The surface formation-center offset is sampled uniformly inside the 800 km disk around the initial TC center; both bearing and radial distance are randomized. This models the assumed upper bound of TC track-forecast error.
Surface validation center offsets	R_{test}	100/200/400/600/800 km	Fixed surface formation-center offsets with random bearings, evaluated without retraining. These values represent increasing track-forecast displacement.
Underwater initialization radius	$R_{0,\text{u}}$	$\leq$200 km	After surface convergence, subsurface vehicles are initialized uniformly within 200 km of the TC center. The 0–800 km forecast-error protocol is surface-only.
Decision interval	Δt	10 min	Environmental observation and action interval.
Surface formation radius	R_{ring}	100 km	Surface SUVs use equal angular spacing of $360°/N$ on a 100 km ring.
Surface fleet sizes	N	4/8/12/16	Number of SUVs in the final surface configurations.
Underwater fleet size	N	4	Number of SUVs in each active-IPPO, random-control, and passive-drift case.
Underwater depths below the sea surface	z	0/10/20/30/40/50/60 m	Seven ocean-current observation depths used in the final subsurface matrix.
Underwater methods	–	Active IPPO; random; drift	Three paired methods evaluated on the same episodes and deployments.
Underwater duration	H	20 h	Fixed mission window for the final subsurface comparison.
Surface speed limits	$V_{\max}^{\text{s}}$	12 or 20 m/s	Two surface-speed scenarios.
Submerged speed limits	$V_{\max}^{\text{u}}$	1.5 or 2.0 m/s	Two resultant propulsion-speed limits after unit-disk action projection, used for the active-control comparison.
Compute device	–	CPU	All reported PPO/IPPO experiments were trained on CPU.

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